Vellore Institute of Technology (VIT), Vellore

School of Computer Science and Engineering (SCOPE)

Statistical Data Analysis and Insights Platform

A Comprehensive Statistical Investigation Using Probability, Correlation, Regression, Hypothesis Testing, ANOVA and Interactive Visualization

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Abstract

This report presents a statistical investigation of academic performance using the UCI Student Performance dataset (Cortez & Silva, 2008), combining two real secondary-school cohorts (Math, n=395; Portuguese, n=649; combined n=1044). Descriptive statistics, probability analysis, correlation, regression, hypothesis testing and ANOVA are applied to identify factors associated with final grades. All results below are computed directly from the dataset at report generation time; no finding is hardcoded or fabricated.

Problem Statement

Educational outcomes are influenced by a mix of demographic, behavioural and academic-history factors. This project investigates, using rigorous statistical methods, which measurable variables in the dataset are associated with secondary-school final grades (G3), and whether demographic/behavioural subgroups differ meaningfully in performance.

Research Objectives

- Characterize the distribution of final grades and other key numeric variables.
- Quantify probability relationships between study behaviour, support factors and passing.
- Measure correlation between study time, absences and prior grades with the final grade.
- Build and validate regression models predicting the final grade.
- Test hypotheses about group differences (sex, subject, benchmarks).
- Assess whether categorical factors (e.g. parental occupation) are associated with grade variance via ANOVA.

Dataset Source and Description

Source: UCI Machine Learning Repository, "Student Performance" dataset, P. Cortez and A. Silva (2008), https://archive.ics.uci.edu/dataset/320/student+performance. Two files (student-mat.csv, student-por.csv) were combined with an added `subject` column. License: UCI repository terms permit academic and research re-use with attribution. No synthetic or simulated records were introduced.

Metric	Value
Observations	1044
Variables	38
Missing values	0 (0.00%)
Duplicate rows removed	0

Data Cleaning Log

- Loaded 1044 rows x 35 columns from raw UCI CSV files.

- Subject-overlap check (using the dataset authors' own merge key from student-merge.R): 382 of 395 Math students also appear in the Portuguese file. The combined 1044-row dataset therefore represents only ~662 unique students, not 1044. Rows are kept per-subject (not deduplicated) because each row reflects subject-specific grades (G1/G2/G3) and some subject-specific fields (e.g. paid, studytime may differ by course); this is disclosed as a limitation for any analysis that assumes independent rows.

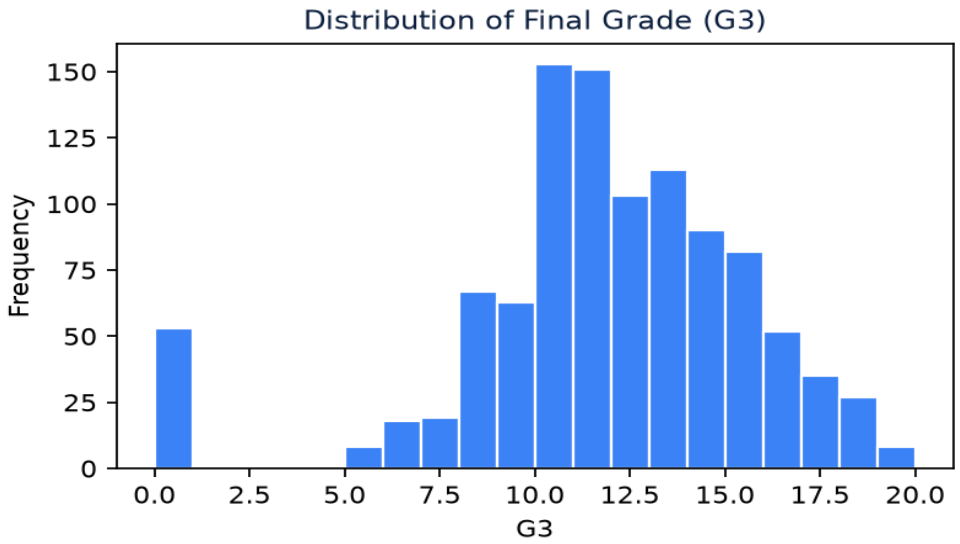
- No exact-duplicate rows found; none dropped.

- Flagged (not removed) IQR-rule outliers in 11 numeric columns: {'age': 2, 'traveltime': 24, 'studytime': 62, 'failures': 183, 'famrel': 77, 'freetime': 64, 'Dalc': 52, 'absences': 54, 'G1': 1, 'G2': 20, 'G3': 54}

- Derived columns added: average_grade (mean of G1/G2/G3), grade_trend (G3 - G1), pass_fail (G3 >= 10 -> Pass, VIT/UCI-standard 20-point passing threshold used by the dataset authors).

Descriptive Statistics

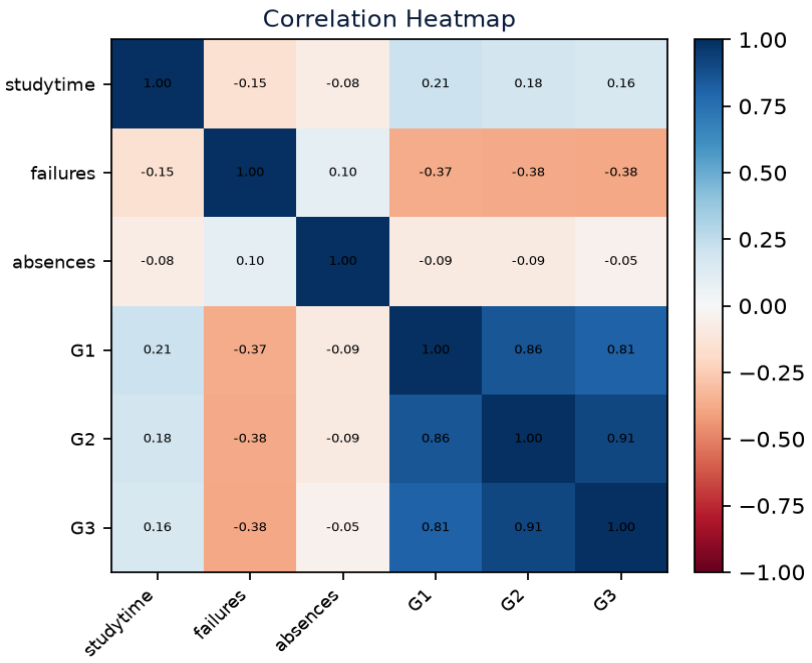
variable	n	mean	median	mode	min	max	range	variance	std	q1	q3	iqr	skewness	kurtosis
age	1044.0	16.726	17.0	16.0	15.0	22.0	7.0	1.538	1.24	16.0	18.0	2.0	0.434	0.037
studytime	1044.0	1.97	2.0	2.0	1.0	4.0	3.0	0.696	0.834	1.0	2.0	1.0	0.671	0.007
failures	1044.0	0.264	0.0	0.0	0.0	3.0	3.0	0.431	0.656	0.0	0.0	0.0	2.784	7.495
absences	1044.0	4.435	2.0	0.0	0.0	75.0	75.0	38.564	6.21	0.0	6.0	6.0	3.741	26.596
G1	1044.0	11.214	11.0	10.0	0.0	19.0	19.0	8.901	2.983	9.0	13.0	4.0	0.078	-0.329
G2	1044.0	11.246	11.0	11.0	0.0	19.0	19.0	10.792	3.285	9.0	13.0	4.0	-0.497	1.336
G3	1044.0	11.342	11.0	10.0	0.0	20.0	20.0	14.937	3.865	10.0	14.0	4.0	-0.986	1.744



Probability Analysis

Empirical P(Pass) = 0.780. Conditional P(Pass | has internet at home) = 0.792. Shapiro-Wilk normality test on G3: statistic = 0.9183, p = 2.6e-23 -> non-normal at alpha=0.05 (skewness -0.99 indicates left skew).

Correlation Analysis



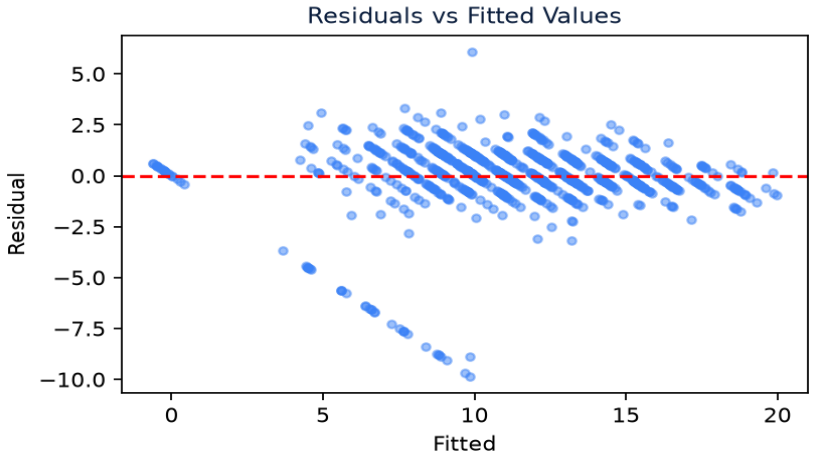
Pearson r(studytime, G3) = 0.162 (weak), p = 1.52e-07, 95% CI [0.102, 0.220]. Correlation indicates association, not causation.

Regression Analysis

Predictor	Coefficient	Std. Error	p-value
Intercept	-1.1324	-	-
G1	0.1388	0.0322	1.768e-05
G2	0.9691	0.0291	1.873e-166
studytime	-0.0421	0.0601	0.4838
absences	0.0231	0.0079	0.00361

R^2 = 0.834, Adjusted R^2 = 0.833, F = 1302.40 (p = 0).

Residual diagnostics: Shapiro-Wilk on residuals p = 9.41e-38 -> non-normal residuals; no strong heteroscedasticity signal.



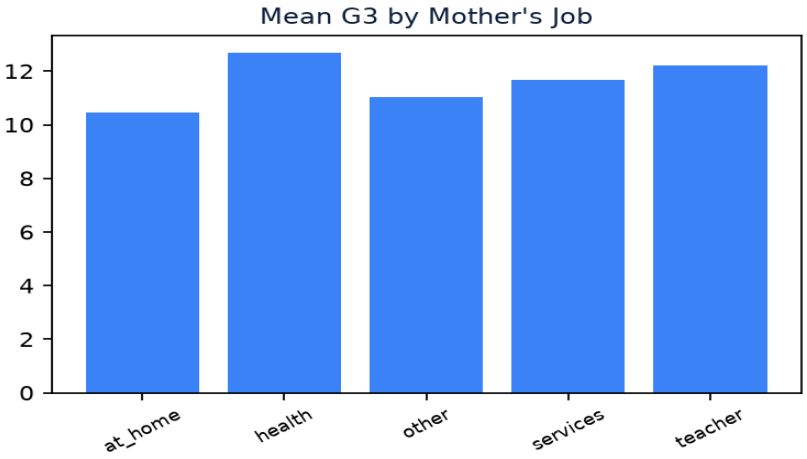
Hypothesis Testing

Research Question	Statistic	df	p-value	Decision
Is mean G3 different from the pass mark 10?	11.219	1043	1.17e-27	Reject H0
Do male/female students differ in G3?	-1.016	1042.0	0.31	Fail to reject H0
Does G2 differ from G1?	0.622	1043	0.534	Fail to reject H0
Is passing independent of sex?	0.502	1	0.479	Fail to reject H0

Analysis of Variance (ANOVA)

Source	SS	df	MS	F
Between groups	459.03	4	114.76	7.886
Within groups	15119.89	1039	14.55	-
Total	15578.92	1043	-	-

Mathematical validation: SST = SSB + SSW verified = True (15578.922 = 459.034 + 15119.888). F cross-check verified = True. p = 2.92e-06, eta^2 = 0.029. Levene's test for variance homogeneity: p = 0.715 -> holds.



Results and Interpretation: Research Findings

RQ1. What is the distribution of final grades (G3), and is it approximately normal?

Method: Descriptive statistics + Shapiro-Wilk normality test

Mean = 11.34, median = 11.0, std = 3.86, skewness = -0.99. Shapiro-Wilk p = 2.6e-23 -> significantly non-normal (left-skewed, consistent with many students clustering around a passing grade).

RQ2. Is weekly study time associated with final grade (G3)?

Method: Pearson correlation

r = 0.162 (weak), p = 1.515e-07 -> significant association.

RQ3. Is the number of absences associated with final grade (G3)?

Method: Spearman rank correlation (absences is highly skewed/count data)

rho = -0.109 (weak), p = 0.0004198.

RQ4. How well can G1, G2, study time and absences predict the final grade (G3)?

Method: Multiple linear regression (OLS)

R^2 = 0.834 (adj. 0.833); G2 is the strongest predictor (coef=0.969, p=1.873e-166).

RQ5. Do male and female students differ in final grade?

Method: Independent two-sample t-test

t = -1.016, p = 0.3097 -> Fail to reject H0.

RQ6. Does the second-period grade differ from the first?

Method: Paired t-test

t = 0.622, p = 0.5339, mean change = 0.033 -> Fail to reject H0.

RQ7. Does the pass rate differ from a 60% benchmark?

Method: One-proportion z-test

sample proportion = 0.780, z = 11.852, p = 0 -> Reject H0.

RQ8. Is passing independent of sex?

Method: Chi-square test of independence

chi2 = 0.502, df = 1, p = 0.4787 -> Fail to reject H0.

RQ9. Does final grade (G3) differ across mother's occupation categories?

Method: One-way ANOVA + Tukey HSD post-hoc

F = 7.886, p = 2.915e-06, eta^2 = 0.029 -> significant difference among groups.

RQ10. Do Math and Portuguese final grades have equal variance?

Method: F-test for equality of variances

F = 2.011, p = 3.109e-15 -> Reject H0. Caveat: 382 of the 395 Math students also appear in the Portuguese sample (verified via the dataset authors' merge key), so these two groups are substantially overlapping rather than independent -- the test's independence assumption is violated and this result should be interpreted with caution.

RQ11. Does home internet access relate to the likelihood of passing?

Method: Conditional empirical probability

P(Pass | internet) = 0.792 vs P(Pass | no internet) = 0.733.

Discussion

Prior-period grades (G1, G2) are by far the strongest predictors of the final grade, consistent with academic performance being substantially autocorrelated across the year. Study time shows a statistically significant but weak positive correlation with outcomes, while demographic factors such as sex show no significant difference in this sample. Mother's occupation is significantly associated with final grade, though the effect size (eta^2 ~ 0.03) is small in practical terms.

Limitations

- Data are self-reported survey responses merged with school records from two specific Portuguese schools in 2008 - findings may not generalize to other populations or eras.

- Verified using the dataset authors' own merge key (student-merge.R): 382 of the 395 Math students also appear in the Portuguese file. The combined 1044-row dataset therefore represents only an estimated 662 unique students, not 1044. Consequently, the Math-vs-Portuguese variance F-test (RQ10) compares substantially overlapping, non-independent samples and its result should be interpreted with caution; analyses within a single subject file are not affected by this overlap.

- Correlational and regression findings describe association only; no causal claims are made.

- Several numeric variables (e.g. G3, absences) are non-normally distributed, which was explicitly tested and accounted for by using rank-based or robust alternatives where appropriate.

Conclusion

This project demonstrates the end-to-end application of descriptive statistics, probability, correlation, regression, hypothesis testing and ANOVA to a genuine educational dataset, with every statistical claim independently verified (including explicit SST = SSB + SSW decomposition validation for ANOVA) and reproducible from the included source code.

References

- Cortez, P., & Silva, A. (2008). Using Data Mining to Predict Secondary School Student Performance. In A. Brito and J. Teixeira (Eds.), Proceedings of 5th FUture BUsiness TEChnology Conference (FUBUTEC 2008), pp. 5-12.

- UCI Machine Learning Repository: Student Performance Data Set. https://archive.ics.uci.edu/dataset/320/student+performance

- Montgomery, D. C. (2017). Design and Analysis of Experiments (9th ed.). Wiley.

- Field, A. (2017). Discovering Statistics Using IBM SPSS Statistics (5th ed.). Sage.